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Windows Server vs Ubuntu Server vs Rocky Linux

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial Microsoft server OS; evaluation versions are available	Open-source Linux distribution	Open-source Linux distribution
User Interface	Can provide a graphical Desktop Experience or command-line/server-focused installation	Commonly administered through command line	Commonly administered through command line
Package Management	Windows tools and package/application installers	APT / DEB packages	DNF / RPM packages
Security	Includes Windows security features, permissions, firewall, updates, and enterprise security integration	Linux permissions, firewall tools, security updates, SSH and other Linux security mechanisms	Enterprise Linux security features, permissions, SELinux, firewall and security updates
Performance	Depends on installed roles, configuration and hardware	Often suitable for lightweight/headless server deployments	Designed for stable enterprise Linux environments
Typical Use Cases	Active Directory, Windows-based applications, file services and Microsoft-oriented infrastructure	Web servers, cloud workloads, databases, containers and general Linux services	Enterprise servers, web/database services and environments seeking RHEL compatibility
Advantages	Familiar GUI options and strong Microsoft ecosystem integration	Free/open source, large community and extensive server/cloud ecosystem	Free/open source and designed for enterprise/RHEL compatibility
Disadvantages	Licensing can add cost and some deployments require more resources	Command-line administration may have a learning curve for new users	Can also require Linux command-line knowledge and may be less familiar to beginners

Windows Server, Ubuntu Server, and Rocky Linux are operating systems that can be used to provide services in enterprise environments. Although all three can function as server operating systems, they have different approaches to licensing, administration, package management, security, and typical use cases.

Windows Server is developed by Microsoft and is commonly used in organizations that rely heavily on Microsoft technologies. One advantage is that administrators can use graphical management tools, especially when the Desktop Experience installation is used. This can make some administrative tasks easier for students and administrators who are already familiar with Windows. Windows Server is commonly associated with services such as Active Directory, file services, and other Microsoft-based enterprise applications. One disadvantage is that organizations may need to consider licensing costs.

Ubuntu Server is an open-source Linux server operating system. It is commonly administered through the command line and uses APT for package management. Ubuntu Server is widely applicable to web hosting, databases, cloud computing, containers, and other network services. During this activity, I learned that Ubuntu Server can be configured without a full graphical desktop, which allows the administrator to focus on server services and command-line management. However, beginners may initially find Linux commands more difficult than a graphical interface.

Rocky Linux is another open-source Linux distribution designed for enterprise use and compatibility with the Red Hat Enterprise Linux ecosystem. It uses RPM packages and DNF as its package manager. This makes Rocky Linux useful for organizations that want an enterprise-focused Linux environment without using Windows Server. Like Ubuntu Server, administrators should be comfortable working with Linux concepts and command-line tools.

The best operating system therefore depends on the needs of an organization. Windows Server can be appropriate when a company depends on Microsoft services and applications. Ubuntu Server is a strong option for general Linux server workloads, web services, cloud environments, and development. Rocky Linux is useful for organizations that prefer an enterprise Linux platform closely aligned with the RHEL ecosystem.

Overall, this comparison helped me understand that there is no single server operating system that is best for every situation. A system administrator should consider the required applications, licensing, security, available hardware, staff experience, and the services that the server needs to provide before selecting an operating system.